1. What is the purpose of the Software Development Life Cycle (SDLC)?
   A. To design user interfaces appropriate for software
   B. To provide a structured approach to software development
   C. To manage project finances in software development
   D. To create marketing strategies in software development

2. What is the primary goal of the Planning Phase in Systems Development?
   A. Current system evaluation
   B. Project scope definition
   C. Interface design
   D. Requirements gathering

3. Which of the following is an example of an Educational Technology Context?
   A. Financial Learning systems
   B. Supply chain management
   C. Learning Management Systems
   D. Learning Relationship Management

4. Which phase of the custom software development process involves understanding the client's requirements and objectives?
   A. Testing and requirement analysis.
   B. Planning requirement design.
   C. Requirements gathering and analysis.
   D. Requirement gathering Development.

5. What is a disadvantage of a structured system development life cycle?
   A. Lack of documentation
   B. Limited flexibility
   C. Faster development
   D. Increased adaptability

6. What do agile methodologies, such as scrum or kanban, often promote during software development?
   A. Rigid and sequential development.
   B. Collaboration, flexibility, and iterative development.
   C. Standalone prototyping.
   D. Complete methodology for prototyping.

7. Which of the following is an approach to trying out features in the context of a full development methodology?
   A. Prototyping.
   B. Waterfall development.
   C. Incremental development.
   D. Spiral development.

8. Which one of this is a characteristic of the Waterfall Model?
   A. Iterative development
   B. Continuous feedback
   C. Phases overlap
   D. Sequential phases

9. In which model are development and testing performed concurrently?
   A. Incremental Model
   B. RAD Model
   C. V-Model
   D. Spiral Model

10. Which SDLC methodology is known for its flexibility and adaptability, with iterations and frequent feedback?
   A. Waterfall Model
   B. Agile Model
   C. V-Model
   D. Spiral Model

11. What is the main objective of Rapid Application Development (RAD)?
   A. To deliver a high-quality system approach at a high investment cost
   B. To follow a sequential quality system development approach
   C. To deliver a high-quality system at a relatively low investment cost
   D. To discourage user involvement quality system investment cost

12. What is the primary objective of incremental development?
   A. To reduce inherent project risk by breaking a project into smaller segments
   B. To encourage extensive up-front planning
   C. To deliver a high-quality system at a high investment cost
   D. To follow a waterfall-based methodology

13. What is the difference between functional and non-functional requirements?
   A. Functional requirements focus on system performance, while non-functional focus on features.
   B. Functional requirements describe what the system does, while non-functional describe how well it does it.
   C. Functional requirements are optional, while non-functional ones are mandatory.
   D. Functional requirements pertain to hardware, while non-functional pertain to software.

14. Which type of UML diagram is used for representing the structure of a system?
   A. Context Models
   B. Interaction Models
   C. Structural Models
   D. Behavioral Models

15. What is the main difference between the Agile model and traditional software development approaches?
   A. The Agile model discourages user involvement, while traditional approaches encourage it
   B. The Agile model advocates a heavier and more process-centric viewpoint than traditional approaches
   C. The Agile model uses iterative development as a basis, while traditional approaches discourage it
   D. The Agile model uses linear development as a basis, while traditional approaches discourage it

16. Which model focuses on risk assessment and on minimizing project risk by breaking a project into smaller segments?
   A. Waterfall model
   B. Incremental funding methodology
   C. Chaos model
   D. Spiral development

17. What is the purpose of the Software Development Life Cycle (SDLC)?
   A. To design user interfaces
   B. To provide a structured approach to software development
   C. To manage project finances
   D. To create marketing strategies

18. What is the primary purpose of storyboarding in educational application development?
   a. To serve as a visual roadmap for planning and organizing the project flow
   b. To replace technical documentation planning and organizing the project flow
   c. To finalize the user interface design planning and organizing the project flow
   d. To test the application's performance planning and organizing the project flow

19. Which UML diagram is used for representing the structure of a system?
   A. Interaction Models
   B. Behavioral Models
   C. Context Models
   D. Structural Models

20. What is the main purpose of programming languages in software development?
   A. Creating user interfaces
   B. Designing hardware components
   C. Communicating with computers
   D. Managing project finances

21. What is the primary focus of non-functional requirements in system development?
   A. Describing system features and its components functions
   B. Outlining system performs **its** behavior functions
   C. Defining the system's look and feel
   D. Describing how well the system performs its functions

22. Which of the following is a characteristic of the Agile development model?
   A. Minimal customer involvement
   B. Fixed project requirements
   C. Emphasis on comprehensive documentation
   D. Adaptive to changing requirements

23. What is the purpose of the Feasibility Study in software application development?
   A. To create user interfaces
   B. To evaluate the technical aspects of the system
   C. To determine if the project can be developed within budget
   D. To manage project finances

24. What is an essential step in software deployment?
   A. Ignoring user feedback
   B. Developing new features of post-deployment
   C. Establishing an Operational Support Plan
   D. Reducing testing efforts

25. It Is critical to test an application before making it available to users.
   A. TRUE
   B. FALSE
   C. Not applicable

SECTION B

[75 Marks]

Answer **only three** questions in this section

1. Explain any 5 phases of the Software Development Life Cycle (SDLC) and their significance in the development process.[25 Marks]

2. Compare and contrast the Waterfall and Agile SDLC methodologies, highlighting their advantages and disadvantages. [25 Marks]

3. Why is it critical to test an application before making it available to users? State any 5 reasons.

4. Explain the following terms (25 MARKS)
   - Alpha testing
   - Beta Testing
   - UML
   - Prototype
   - Hyperlink

5. a) Explore the role of programming languages in system development, highlighting the differences between low-level and high-level languages.

   b) Investigate the history and benefits of Integrated Development Environments (IDEs), and discuss their role in modern software development.